Documentation for the quadrilateral certificates

Supplement to *Upper bounds for packings of cylinders around the unit sphere*

October 30, 2026

This package verifies the quadrilateral bounds for every $4 \le N \le 24$. It contains the rational parameters, finite subdivisions, and Python proof checkers. No optimization or search program is needed to check the results.

1 Running and reading the certificates

Use Python 3.10 or later, with the standard library only. From the extracted package directory, run

```
python run_verification.py 7
python run_verification.py --jobs 4
```

The first command checks the seven-cylinder bound. The second checks all 21 cases, with four independent case checks at a time. Omitting `--jobs` uses one process at a time. Runs can be lengthy.

The runner prints `PASS N=...` only after the corresponding complete case finishes. The final success message requires every requested case to pass. A report must have `shard: null`, no failures, and `checked_boxes = overlap_boxes`; a partial shard is not a certificate. The runner also checks these conditions. Do not run with `-O`, `-OO`, or a nonempty `PYTHONOPTIMIZE` setting: assertions perform proof checks, and the release refuses optimized execution.

Each family has three source modules:

- `verify_bounds.py`: parameters, scalar inequalities, subdivision coverage, and the full verification loop;
- `exact_geometry.py`: interval arithmetic, affine clipping, spherical area bounds, and box reconstruction;
- `complement_geometry.py`: connectedness of the directed difference.

The directories `bounds/` and `bounds_13_24/` cover $4 \le N \le 12$ and $13 \le N \le 24$, respectively. The latter uses a more general enclosure of sinusoidal coefficient extrema. The geometric proof, the three properties checked, and the data format are the same.

2 Parameters and the claims certified

For each N, `data/nN_case.json` stores decimal strings interpreted as exact rational numbers:

$$N, \quad r, \quad u, \quad v, \quad \rho, \quad \delta, \quad \lambda.$$

The JSON keys for ρ and λ are `R` and `angle_lower`. The case also names two compressed subdivision files: `nN_overlap.json.gz` and `nN_cutoff.json.gz`. The historical name `cutoff` now

means a connected-difference certificate: every accepted box has all four pieces marked. No sliver is discarded, and there is no extra area penalty.

Put $H = 1 + r$ and take the reference axis $L_1 = (0, -H, 0) + \mathbb{R}(0, 0, 1)$. The inverse gnomonic map

$$g(X, Z) = \frac{(X, -1, Z)}{\sqrt{1 + X^2 + Z^2}}, \qquad D = [-u, u] \times [-v, v],$$

defines the quadrilateral $Q = g(D)$. For all admissible cylinder pairs and triples, the checks establish:

1. no common point of three assigned closed quadrilaterals;
2. connectedness of each directed difference $Q_i \setminus Q_j$;
3. $|Q_i \cap Q_j| \le \delta$.

They also prove, using a rational lower bound for $A = 4 \arctan(uv/\sqrt{1 + u^2 + v^2})$, that

$$NA - (3N - 6)\delta > 4\pi.$$

The manuscript's geometric theorem then excludes radius r. The programs check its numerical hypotheses; they do not formally verify the topological proof of that theorem.

3 The scalar checks

For three axes separated pairwise by at least $2r$, let a_i be their nearest points to P and $\bar{a} = (a_1 + a_2 + a_3)/3$. Then

$$12r^2 \le \sum_{i<j} |a_i - a_j|^2 = 3 \sum_i |a_i - \bar{a}|^2 \le 3 \sum_i |a_i - P|^2.$$

Thus no point is strictly closer than $2r/\sqrt{3}$ to all three axes. Write $S = 1 + u^2$, $T = S + v^2$. The largest squared distance from ρQ to L_1 is

$$q^2 = \max\left\{ \rho^2 - \frac{2H\rho}{\sqrt{S}} + H^2,\ \frac{\rho^2 S}{T} - \frac{2H\rho}{\sqrt{T}} + H^2 \right\}.$$

Indeed, the squared distance increases with X^2 for fixed Z; after setting $X^2 = u^2$, it is a convex quadratic in $(S + Z^2)^{-1/2}$, so its maximum occurs at $Z = 0$ or $|Z| = v$. The checker verifies $q^2 < 4r^2/3$ by checking positive signs and squaring the two inequalities. A common spherical point would yield a forbidden point after multiplication by ρ.

The additional rational inequality

$$|\rho - H| + \frac{7}{6} r < 2r$$

ensures that each tangency point c_i belongs to no other Q_j: otherwise ρc_i is at distance $|\rho - H|$ from L_i and at most $q < 7r/6$ from L_j, contradicting their separation. The connectivity check separately certifies a uniform initial mark in each retained parameter box.

4 The parameter box and the frame convention

The following explicit frame removes any ambiguity about the angle w. Put

$$s = \sin\theta, \quad c = \cos\theta, \quad t = \sin\phi, \quad d = \cos\phi,$$

and define

$$\begin{aligned}\nu &= (c, s, 0),\\ \eta &= (s\cos w, -c\cos w, -\sin w),\\ c_2 &= t\nu - d\eta, \qquad e_2 = d\nu + t\eta,\\ a_2 &= (-s\sin w, c\sin w, -\cos w).\end{aligned}$$

The second cylinder has axis $L_2 = Hc_2 + \mathbb{R}a_2$; c_2 is its point of tangency and e_2 is its transverse direction. These vectors form an orthonormal frame. Its four inward normals are

$$uc_2 + e_2, \quad uc_2 - e_2, \quad vc_2 + a_2, \quad vc_2 - a_2.$$

Taking their dot products with $(X, -1, Z)$ produces exactly the four affine inequalities used by the code.

For a nonparallel pair choose ν along its common perpendicular, directed from the first axis to the second. Then $\nu \cdot L_1 = -Hs$, $\nu \cdot L_2 = Ht$, and the separation is $H(s+t)$. A reflection preserving the reference cylinder makes the x-component of ν nonnegative. The transverse direction of the second tangency point around ν determines w; reflection in $z = 0$ allows $0 \le w \le \pi$. Reversing an axis direction does not change its cylinder or quadrilateral. The endpoints include parallel axes. In degenerate cases the frame is not unique, which does not affect coverage. The unsigned angle between the unoriented axes alone is $\arccos|\cos w|$ and should not replace this frame convention.

Every admissible pair has

$$s + t \ge K := \frac{2r}{1+r}, \qquad \theta, \phi \ge \arcsin(K-1).$$

The actual rational root box is

$$[\lambda, P/2]^2 \times [0, P], \qquad P = 3.141592653589794 > \pi.$$

The checker proves $\sin\lambda < K - 1$, with λ in the increasing branch of sine, hence $\lambda < \arcsin(K-1)$. This box contains the physical parameter range. The tiny excess beyond $\pi/2$ and π is retained and handled by the interval bounds.

For overlap area only, exchange of the cylinders permits restriction to $\theta \le \phi$. Both orders remain in the connected-difference check. A box is discarded for insufficient separation only when an upper bound for $\sin\theta + \sin\phi$ is strictly less than K.

5 How a finite subdivision covers all configurations

After decompression, a subdivision is a JSON list of pairs `[path, reason]`. The path is a string of zeros and ones. Start with $[0,1]^3$. At each step bisect the coordinate with largest weighted width, using weights $(1, 1, 2)$; ties go to the first coordinate. A zero selects the lower half and a one the upper half. Finally map the first two coordinates affinely to $[\lambda, P/2]$ and the third to $[0, P]$. All endpoints are rational.

The checker rejects invalid path characters, duplicate or prefix-related leaves, and inappropriate reason labels. If m is the maximum path length, it also checks

$$\sum_{\text{leaves } p} 2^{m-|p|} = 2^m.$$

For binary, prefix-free paths this proves complete coverage of the root box by the dyadic subdivision. Shared box boundaries are included. Allowed reasons are `separation` and `cutoff` for connectivity; and `separation`, `order`, and `overlap` for area. Every reason is recomputed, not trusted. In particular `order` requires the entire box to satisfy $\theta > \phi$.

6 Connected differences: precise meanings of the marks

For a parameter box, enclose the coefficients in

$$\ell_k(X, Z) = a_k X + b_k Z + d_k, \qquad B_k = \{(X, Z) \in D : \ell_k(X, Z) < 0\}.$$

Every nonempty B_k is convex. A mark means: for every configuration in this box, this piece, if nonempty, belongs to the component containing $(0, 0)$. Initialization requires at least one fixed index k with $\ell_k(0, 0) < 0$ throughout the box.

There are two ways to extend the marked union.

1. *Uniform witness.* Find a fixed point $p \in D$ for which upper bounds for both $\ell_i(p)$ and $\ell_j(p)$ are strictly negative, where B_i is marked. Then B_i and B_j intersect for every configuration. The program first tests the nine corners, side midpoints, and center of D. It also seeks a witness by clipping quadrants by conservative half-planes and testing the resulting vertex average.

2. *Potential vertices.* List every vertex v of D for which the coefficient bounds do not rule out $\ell_k(v) < 0$. The program conservatively includes vertices with lower bound ≤ 0. Require every listed vertex to lie in some marked piece throughout the box. Whenever B_k is nonempty, an affine function has a strictly negative minimum at a vertex, and this vertex connects B_k to the marked union. The connecting vertex and marked piece may vary. Empty pieces satisfy the marking condition vacuously.

In particular, the second test does not require a vertex to lie in B_k for the whole box. This is why it can succeed when the first test fails. All final accepted connectivity boxes have all four marks.

7 Overlap area and outward rounding

On a fixed quadrant of D, signs of X, Z are known. The interval endpoints therefore give an affine upper bound $\ell_k^+(X, Z) \ge \ell_k(X, Z)$ throughout the parameter box. Replacing $\ell_k \ge 0$ by $\ell_k^+ \ge 0$ yields a rational enclosing polygon. The code rounds the coefficients further outward to a grid of scale 10^{15}, which can only enlarge that polygon.

Triangulate each enclosing polygon about its vertex average. For rays $a = (X_a, -1, Z_a)$ and similarly b, c in one quadrant, the spherical area of a triangle is

$$2 \arctan \frac{|\det(a, b, c)|}{\|a\|\|b\|\|c\| + (a \cdot b)\|c\| + (b \cdot c)\|a\| + (c \cdot a)\|b\|}.$$

All dot products are positive. Integer square roots and downward rounding bound the denominator below; upward rounding bounds the ratio above. The checker uses $\arctan x \le x$, or $\arctan x \le x - x^3/3 + x^5/5$ when the upper ratio is below one. The latter polynomial is increasing. Summing the outward-rounded bounds over the four quadrants gives a valid upper bound for every configuration in the box. It must not exceed the exact rational δ.

Geometric interval endpoints are stored as integers at scale 10^{40}, and area upper bounds at scale 10^{20}. Sine and cosine are enclosed by Taylor polynomials through degrees 79 and 78, with a proved remainder smaller than 10^{-40} on the required range. Square roots use `math.isqrt`; π is bounded using Machin's identity and alternating arctangent series. Floating-point values appear only in timing and human-readable reports, never in a proof decision.

For $N \le 12$, the checker verifies the sign hypotheses for the sharper monotonicity-based coefficient bounds. For $N \ge 13$, the more negative lower angular endpoints require including all possible extrema of $A\sin\phi + B\cos\phi$. On an interval shorter than π, derivative endpoint signs identify its possible extremum, and the enclosure includes the appropriate value $\pm\sqrt{1+u^2}$. Ambiguous signs enlarge the bound.

8 Additional rational parameters

All numbers in the following table are exact; ρ is the scale used in the no-triple check and λ the lower endpoint of the rational parameter box.

N	ρ	λ
4	3.867351593	0.475
5	2.972446884	0.263
6	2.532925205	0.112
7	2.2660647	-0.002
8	2.087209326	-0.089
9	1.965724423	-0.160
10	1.872956913	-0.220
11	1.798736145	-0.272
12	1.738840113	-0.316
13	1.687090606	-0.357
14	1.644078543	-0.392
15	1.606533575	-0.423
16	1.573465805	-0.453
17	1.545090121	-0.478
18	1.518963900	-0.503
19	1.496107531	-0.525
20	1.476495144	-0.544
21	1.457644638	-0.564
22	1.439988854	-0.582
23	1.424936166	-0.598
24	1.409772981	-0.614